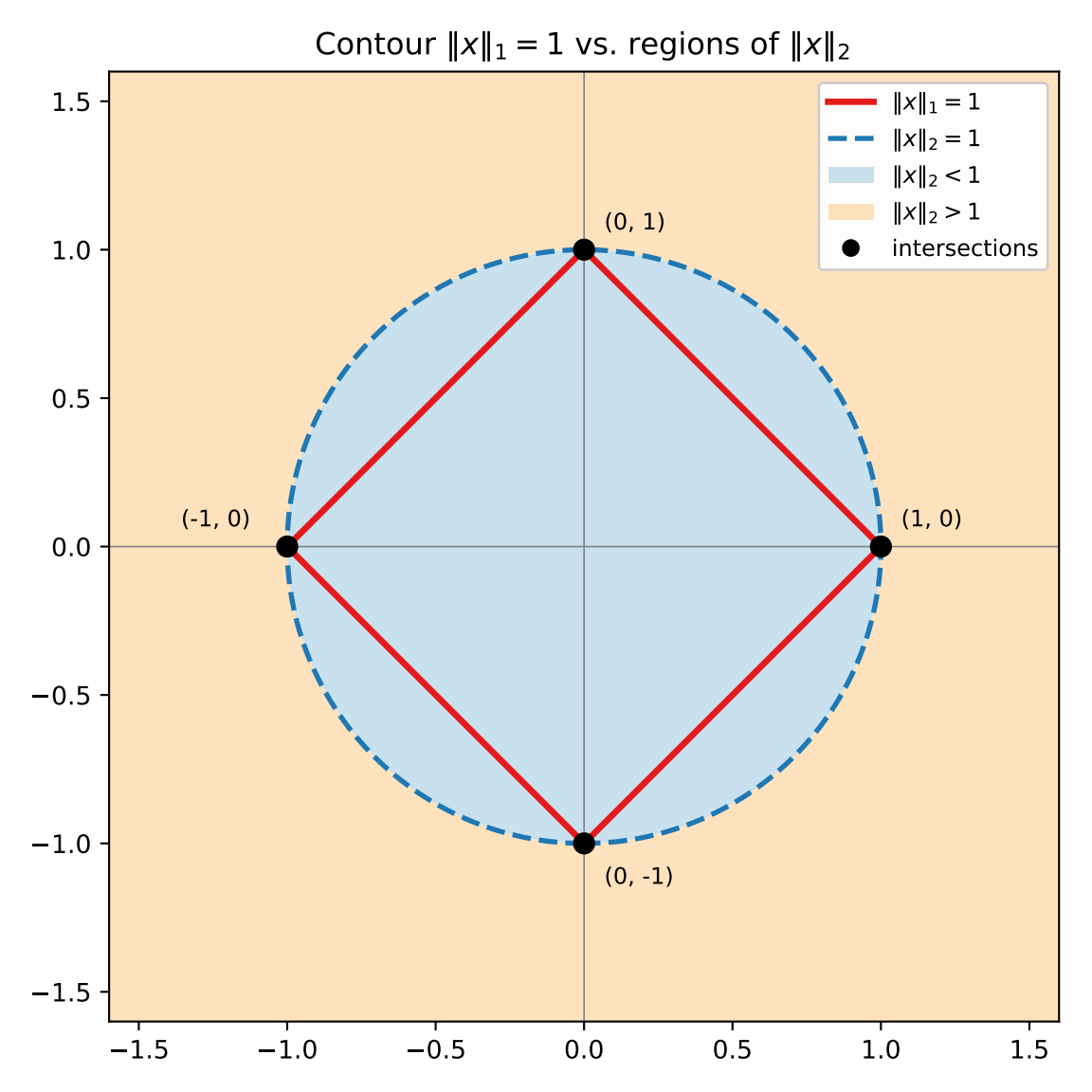

Contour $\|x\|_1 = 1$ vs. regions of $\|x\|_2$
$\|x\|_1 = 1$
$\|x\|_2 = 1$
$\|x\|_2 < 1$
$\|x\|_2 > 1$
intersections
(0, 1)
(-1, 0)
(1, 0)
(0, -1)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5